

WEEK 3 - REPORT

E-Governance Analytics: Visualizing Panchayat
Digital Participation and Governance Indicators

**Topic: Data Visualization and Exploratory Analysis of
E-Governance Indicators**

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Internship name: Yuva intern

Internship Course Name: Virtual Data Analysis Intern –
E-Governance & Digital Services.

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1.Introduction

After completing the cleaning and pre-processing stage in Week 2, the next step was to understand the dataset through visualization and analysis.

Working only with tables and numerical values made comparison difficult, so graphs were used to make governance indicators easier to understand.

This stage mainly focused on comparing state-wise governance performance and observing patterns in digital participation.

2.Objectives of Analysis

The objectives during this stage were:

- To create visual representations of governance indicators
- To compare state-wise governance participation
- To identify trends and patterns from governance data
- To understand relationships between different indicators
- To generate insights from the cleaned dataset

3.Dataset Description

The dataset used for visualization was the cleaned Panchayat governance dataset prepared during Week 2.

The dataset contained indicators related to:

- Development plan uploads
- Online payment participation
- Panchayat onboarding information
- Block and Zilla Panchayat activity

Two additional columns created during pre-processing were also used:

- Total_number(sum)
- Adoption_percentage

Since cleaning and validation had already been completed earlier, the dataset was directly used for analysis.

4.Exploratory Data Analysis

Before creating graphs, the dataset was explored to understand its structure and numerical values.

During this process:

- Dataset preview functions such as `head()` and `info()` were used
- Statistical summaries were checked using `describe()`
- Governance indicators were sorted to identify higher performing states
- Numerical relationships were observed before visualization

These steps helped in deciding which indicators should be visualized.

Dataset Preview:

The screenshot shows a Jupyter Notebook interface with the file explorer on the left. The notebook has two tabs: 'week3_visualization.ipynb' and 'week3_visualization1.ipynb'. The active tab is 'week3_visualization.ipynb', which contains the following code and output:

```
df.head()
```

	States	Gram_Panchayat_Development_Plans_Uploaded	Gram_Panchayats_Ported	Total_Number_of_Block_Panchayats_& Equivale
0	Andhra Pradesh	13371	13351	130
1	Arunachal Pradesh	2108	1577	20
2	Assam	2666	2119	22
3	Bihar	8135	8135	81
4	Chhattisgarh	11658	11658	116

```
df.tail()
```

	States	Gram_Panchayat_Development_Plans_Uploaded	Gram_Panchayats_Ported	Total_Number_of_Block_Panchayats_& Equivale
25	Telangana	12769	12769	
26	Tripura	1219	1178	
27	Uttarakhand	7791	7791	

The screenshot shows a Jupyter Notebook interface with the file explorer on the left. The notebook has two tabs: 'week3_visualization.ipynb' and 'week3_visualization1.ipynb'. The active tab is 'week3_visualization.ipynb', which contains the following code and output:

```
Checking datatype before visualization
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 12 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   States                                                                30 non-null    str
1   Gram_Panchayat_Development_Plans_Uploaded                          30 non-null    int64
2   Gram_Panchayats_Ported                                              30 non-null    int64
3   Total_Number_of_Block_Panchayats_& Equivalent                      30 non-null    int64
4   Block_Panchayats_Plan_uploaded                                     30 non-null    int64
5   Total_Number_of_ZillaPanchayats_& Equivalent                      30 non-null    int64
6   ZillaPanchayats_Plan_uploaded                                       30 non-null    int64
7   Gram_Panchayat_on_onboard                                           30 non-null    int64
8   Gram_Panchayat_With_Online_Payment                                30 non-null    int64
9   Plans_Uploaded                                                       30 non-null    int64
10  Total_number(sum)                                                    30 non-null    int64
11  Adoption_percentage                                                  29 non-null    float64
dtypes: float64(1), int64(10), str(1)
memory usage: 2.9 KB
```


Statistical Summary

The screenshot shows a Jupyter Notebook titled "week3_visualization.ipynb" in a VS Code environment. The notebook is at cell 1, which is titled "IMPORTING LIBRARIES". The code in the cell is:

```
df.describe()
```

The output of the code is a statistical summary of the dataset. The summary is presented as a table with the following columns: **Gram_Panchayat_Development_Plans_Uploaded**, **Gram_Panchayats_Ported**, **Total_Number_of_Block_Panchayats_& Equivalent**, and **Block**. The rows represent various statistical measures: count, mean, std, min, 25%, 50%, 75%, and max.

	Gram_Panchayat_Development_Plans_Uploaded	Gram_Panchayats_Ported	Total_Number_of_Block_Panchayats_& Equivalent	Block
count	30.000000	30.000000	30.000000	
mean	9031.533333	8500.700000	7958.533333	
std	11442.794802	11652.160966	11778.344491	
min	185.000000	0.000000	0.000000	
25%	2247.500000	1290.500000	705.000000	
50%	6095.000000	4606.500000	3981.000000	
75%	12308.250000	12306.750000	11551.250000	
max	58189.000000	58189.000000	58189.000000	

Below the table, the text "Checking datatype before visualization" is displayed. The code in the next cell is:

```
df.info()
```

The output of the code is the datatype of the dataset, which is a pandas DataFrame.

```
<class 'pandas.DataFrame'>
```

5.Visualizations Performed

5.1.Adoption Percentage Analysis

A bar chart was created to compare states based on adoption percentage.

This graph helped in understanding which states showed stronger governance participation.

Observation:

Some states showed significantly higher participation compared to others, indicating differences in governance implementation.

File Edit Selection View Go Run ... yuv internship

EXPLORER

- YUV INTERNSHIP
 - e_governance-dataset.csv
 - e_governance1.ipynb
 - e_governance_research.ipynb
 - e_panchayat_cleaned_v1.csv
 - graph1.png
 - SS1_Cleaned_Dataset_Preview.png
 - SS2_Dataset_Structure.png
 - SS3_Statistical_Summary.png
 - week3_visualization.ipynb
 - week3_visualized_dataset.csv

week3_visualization.ipynb

4.1.TOP 10 STATES BY ADOPTION PERCENTAGE

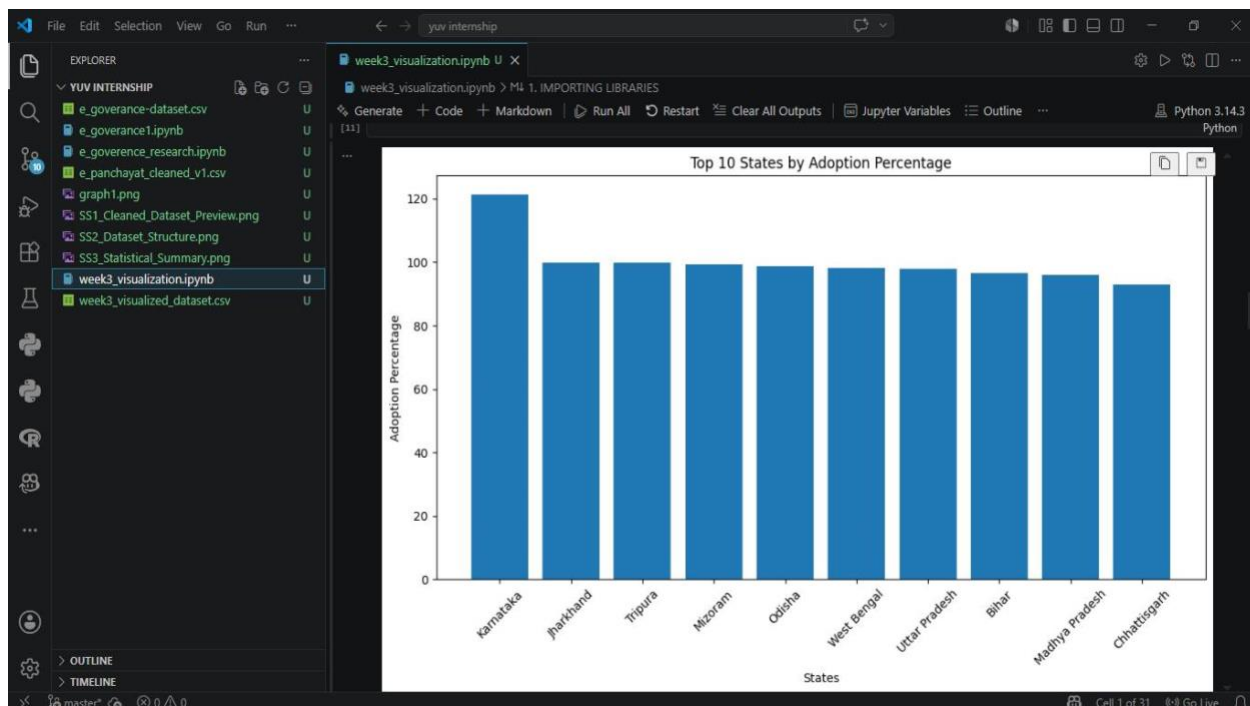
this graph shows the top 10 states by using adoption percentage

```
top_adoption = df.sort_values(
    by='Adoption_percentage',
    ascending=False
).head(10)
```

```
plt.figure(figsize=(12,6))
plt.bar(
    top_adoption['States'],
    top_adoption['Adoption_percentage']
)
plt.xticks(rotation=45)
plt.title("Top 10 States by Adoption Percentage")
plt.xlabel("States")
plt.ylabel("Adoption Percentage")
```

Python

Cell 1 of 31 Go Live



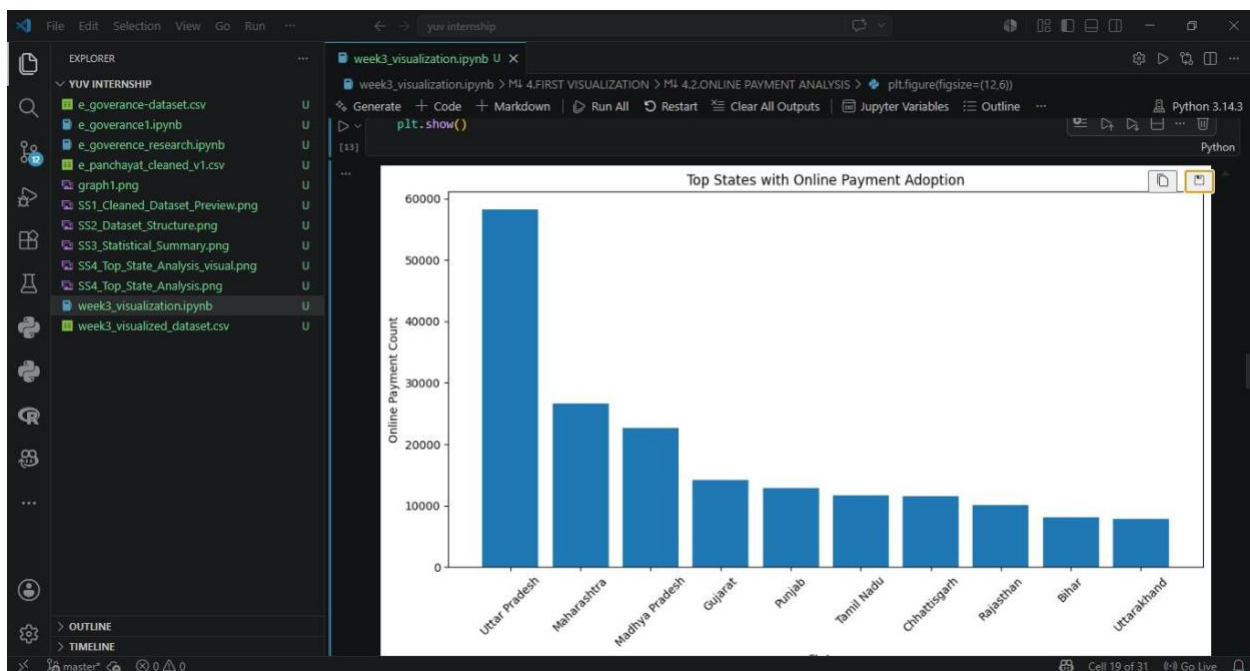
5.2.Online Payment Adoption Analysis

A separate visualization was created for online payment indicators.

This helped compare digital payment participation across states.

Observation:

Online payment adoption was not equally distributed and showed noticeable variation between states.

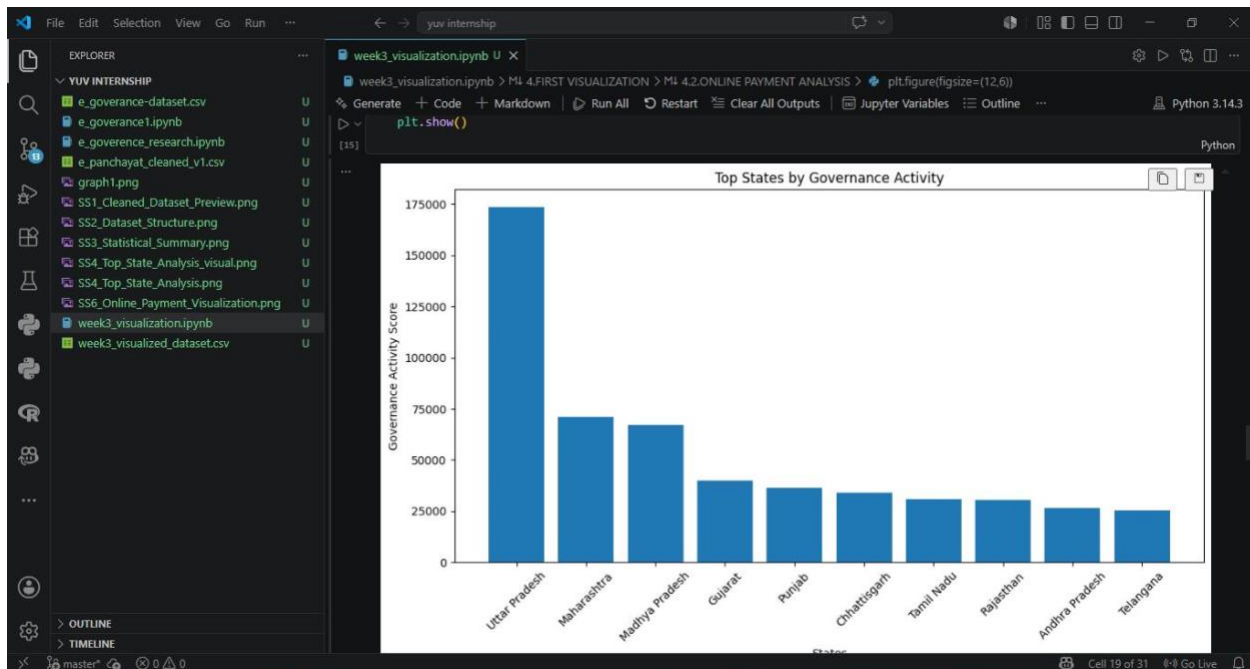


5.3.Governance Activity Analysis

The feature engineered Total_number(sum) column was used to compare governance activity levels.

Observation:

States with higher governance activity generally showed stronger participation across multiple indicators.

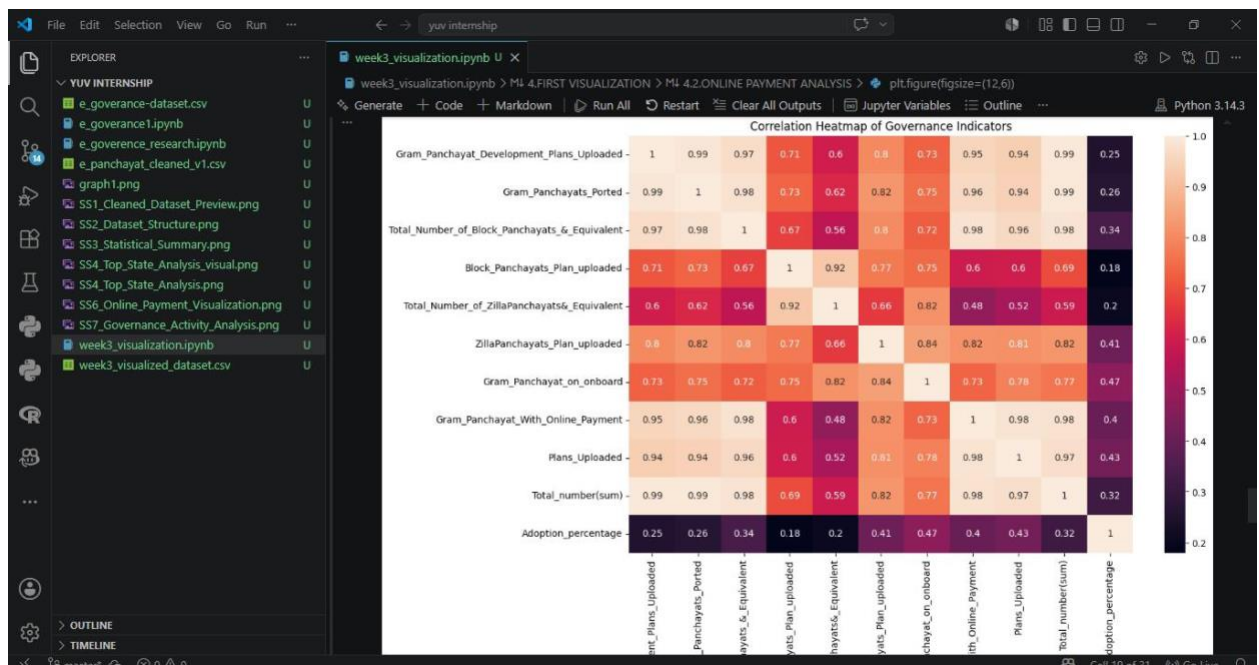


5.4. Correlation Analysis

A heat map was generated to understand relationships between governance indicators.

Observation:

Some indicators showed positive relationships, suggesting that stronger performance in one governance metric often corresponded with higher values in others.



6.Key Findings

Some important findings from the analysis were:

- Governance participation varied considerably across states
- Some states demonstrated stronger digital adoption compared to others
- Online payment participation showed visible differences across regions
- Visualization made state-wise comparison easier
- Additional analytical columns improved comparative analysis

The screenshot displays a Jupyter Notebook environment. On the left, a file explorer shows a project named 'YUV INTERNSHIP' containing various files like 'e_governance-dataset.csv', 'e_governance1.ipynb', and several PNG images. The main area shows a notebook file 'week3_visualization.ipynb' with a code cell containing the command `plt.figure(figsize=(12,6))`. Below the code, a 'Key Insights' section lists four bullet points. Further down, a '7.Conclusion' section contains a paragraph summarizing the visualization process. The bottom status bar indicates 'Cell 19 of 31' and 'Go Live'.

File Edit Selection View Go Run ...

EXPLORER

YUV INTERNSHIP

- e_governance-dataset.csv U
- e_governance1.ipynb U
- e_governance_research.ipynb U
- e_panchayat_cleaned_v1.csv U
- graph1.png U
- SS1_Cleaned_Dataset_Preview.png U
- SS2_Dataset_Structure.png U
- SS3_Statistical_Summary.png U
- SS4_Top_State_Analysis_visual.png U
- SS4_Top_State_Analysis.png U
- SS6_Online_Payment_Visualization.png U
- SS7_Governance_Activity_Analysis.png U
- SS8_Correlation_Heatmap.png U
- week3_visualization.ipynb U
- week3_visualized_dataset.csv U

week3_visualization.ipynb U X

week3_visualization.ipynb > M4 4.FIRST VISUALIZATION > M4 4.2.ONLINE PAYMENT ANALYSIS > `plt.figure(figsize=(12,6))`

Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline ... Python 3.14.3

Key Insights:

- Several states demonstrated strong Panchayat digitization participation.
- Online payment adoption varied significantly across states.
- Governance activity indicators showed positive relationships with planning uploads.
- Data visualization improved comparative understanding of governance performance.

7.Conclusion

The visualization process highlighted variations in Panchayat digitization and governance adoption across Indian states. States with higher governance activity also demonstrated stronger online payment integration and planning participation. The analysis successfully transformed raw governance indicators into interpretable insights through comparative visualizations.

OUTLINE

TIMELINE

master 0 0 0 Cell 19 of 31 Go Live

7.Challenges Faced

During this stage, a few challenges were encountered:

- Selecting suitable graphs for different indicators
- Interpreting large numerical differences between states
- Understanding relationships between multiple variables
- Converting numerical outputs into meaningful observations

These issues were gradually solved through experimentation with graphs and repeated analysis.

8.Conclusion

This stage mainly focused on understanding governance indicators using visualization techniques. Creating graphs and comparisons made the dataset easier to interpret and helped identify differences in governance participation between states.

Overall, the visualization process helped convert cleaned governance data into more understandable insights and comparisons.

9.REFERENCES

1. Ministry of Panchayati Raj, Government of India. E-Panchayat Mission Mode Project (eGramSwaraj). Available at:

<https://egramswaraj.gov.in/>

2. Government of India Open Government Data Platform. Public governance and digital services datasets. Available at:

<https://data.gov.in/>

3. Ministry of Electronics and Information Technology, Government of India. Digital Governance Initiatives and Public Service Delivery Information. Available at: <https://www.meity.gov.in/>

4. My GitHub Week 3 project. Available at: [Rohan's GitHub Link](#)